**Code No: 136FT** **R16**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech III Year II Semester Examinations, July - 2023**
**PRINCIPLES OF COMPUTER COMMUNICATIONS AND NETWORKS**
**(Common to CE, EEE, CSE, IT)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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**PART – A**

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | Define Interoperability. | [2] |
| b) | List the types of Computer networks. | [3] |
| c) | Write a short note on OSI reference model. | [2] |
| d) | List out Communication Service Methods. | [3] |
| e) | What are Analog Signals? | [2] |
| f) | Write a short note on Digital Carrier Systems. | [3] |
| g) | Write a brief note on Physical Layer. | [2] |
| h) | Explain briefly Physical characteristics of Wire. | [3] |
| i) | Discuss briefly about PC Cards. | [2] |
| j) | Write a short note on Bridges. | [3] |

**PART – B**

**(50 Marks)**

| | | |
|---|---|---|
| 2. | Write a detailed note on Computer Communications and Networking. | [10] |
| | **OR** | |
| 3.a) | Discuss in detail about Network Addressing. | |
| b) | Briefly explain about Routing and Reliability. | [5+5] |
| 4.a) | Explain various Data Transmission Modes. | |
| b) | Enumerate Computer Applications and Application Protocols. | [5+5] |
| | **OR** | |
| 5.a) | Explain in detail Speed and capacity of a Communication Channel. | |
| b) | Discuss in detail about Computer Communications and Networking models. | [5+5] |
| 6. | Write a detailed note on representing Data as Digital Signals. | [10] |
| | **OR** | |
| 7. | Discuss in detail about Data Rate and Bandwidth Reduction. | [10] |
| 8.a) | Write a detail note on Data Link Layer. | |
| b) | Explain in detail about Wireless Communication. | [5+5] |
| | **OR** | |
| 9. | Write a detailed note on Copper Media and Fiber Optic Media. | [10] |

10.a) Discuss in detail about Media Convertors.
b) Distinguish between switches Vs Routers. [5+5]

**OR**

11.a) Explain in detail about Connectors.
b) Write a detailed note on Repeaters. [5+5]

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